IOT BASED WATER MONITORING SYSTEM

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Project Report

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FOREWARD

We are delighted to write this foreword, Prof. Jayjeet Sarkar has been a good project guide and a good professor for more than three years. We believe deeply in the educative value of interpretive discussion for all students, especially in a democratic society. I also believe that teachers at every level and stage of their career can enrich and strengthen their innovation by the discussion about this project. Leading patterns and practices presented in this project. Participating in interpretive discussions can help all the commercialist in the world alike learn to use their minds with power through this very project. Prof. Jayjeet Sarkar introduced me to the concept of interpretive discussion when we worked together in this very project, he taught us elementary things for this project how to prepare for the prototype and lead interpretive discussions for the decision makings and forwarding of this project. First with his peers and then with our group of four students in their practicum. So, we are really grateful and thankful to you for your precious guidance to make this project fruitful and we get pleasure to work under you. We are very delighted to dedicate this project to you.

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I would like to express my heartiest gratitude to all the professor of the department of Applied Electronics and Instrumentation Engineering, and also the other professors and facilities of other departments of Academy of Technology for enabling me to do this project successfully. Appreciation of kindness to Prof. Hiranmoy Mandal (HOD of Applied Electronics and Instrumentation Engineering), Prof. Samarjit Maity (Assistant professor of Applied Electronics and Instrumentation Engineering), Prof. Basab Chatterjee (Assistant Professor), Prof. Jayjeet Sarkar (Assistant professor of Applied Electronics and Instrumentation Engineering and project guide), Prof. Niloy Ghosh (HOD of Mechanical engineering), Prof. Abhijit Mallik (HOD of Chemistry department) and many other faculties like Saptarshi sir, Chandasree madam & all the other personnel who were directly or indirectly associated with this project for supporting us continuously with their best effort. I am fortunate enough.

I am fortunate enough to pursue an innovative project under the precious guidance of Prof. Hiranmoy Mandal and other professors. The experience gained during this period gave us an insight as to how the processes actually occur in practice.

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CONTENTS

CHAPTER 1

INTRODUCTION

Currently drinking water is very prized for all the humans. In recent times water levels are very low and water in the lakes are going down. So it is too important to find the solution for water monitoring & control system. IoT is a solution. In recent days, development in computing and electronics technologies have triggered Internet of Things technology. Internet of Things can be described as the network of electronics devices communicating among them by the help of a controller. The IoT is a collection of devices that work together in order to serve human tasks in an efficient manner. It combine computational power to send data about the environments. These devices can be in form of sensors, appliances, embedded systems, and data analysis microchips. This paper present a low cost water monitoring system, which is a solution for the water wastage and water quality.

The Smart Water Quality Monitoring System will measure the following water parameters for analysis; Potential Hydrogen (pH), Oxidation and Reduction Potential (ORP), Conductivity and Temperature using a RS technology. While monitoring these parameters, it is perceived that one should receive a stable set of results. Therefore a continuous series of anomalous measurements would indicate the potential introduction of a water pollutant and the user will be notified of this activity with the aid of IoT technology. Hence, with the successful implementation of this monitoring approach, a water pollution early warning system can be achieved with a fully realized system utilising multiple monitoring stations.

Traditionally, detection of water quality was manually performed where water samples were obtained and sent for examination to the laboratories which is time taking process, cost and human resources (Das & Jain, 2017) (He & Zhang., 2012). Such techniques do not provide data in real-time. The proposed water quality monitoring system is consisting of a microcontroller and basic sensors, is compact and is very useful for pH, turbidity, water level detection, temperature and humidity of the atmosphere, continuous and real-time data sending via wireless technology to the monitoring station (Sugapriyaa et al., 2018) (Barabde & Danve., 2015).

Fig 1.1: Pictorial representation of water monitoring system

CHAPTER 2

LITERATURE SURVEY

Wireless sensor networks are also known as wireless sensor and actuator network(WSAN) which is a network consisting of distributed sensors to monitor physical or environmental conditions such as pressure, sound, temperature etc. This system includes a gateway that provides connectivity to the used world and distributed nodes, which can transfer the data through the network to main location. The modern networks are bidirectional in nature and enable the sensor activity.

Nikhil Kedia entitled "Water Quality Monitoring for Rural Areas-A Sensor Cloud Based Economical Project." Published in 2015 1st International Conference on Next Generation Computing Technologies (NGCT-2015) Dehradun, India. This paper highlights theentire water quality monitoring methods, sensors, embedded design, and information dissipation procedure, role of government, network operator and villagers in ensuring proper information dissipation. It also explores the Sensor Cloud domain. While automatically improving the water quality is not feasible at this point, efficient use of technology and economic practices can help improve water quality and awareness among people.[1]

Jayti Bhatt,Jignesh Patoliya entitled "Real Time Water Quality Monitoring System".This paper describes to ensure the safe supply of drinking water the quality should be monitored in real time for that purpose new approach IOT (Internet of Things) based water quality monitoring has been proposed. In this paper, we present the design of IOT based water quality monitoring system that monitor the quality of water in real time. This system consists some sensors which measure the water quality parameter such as pH, turbidity, conductivity, dissolved oxygen, temperature. The measured values from the sensors are processed by microcontroller and this processed values are transmitted remotely to the core controller that is raspberry pi using Zigbee protocol. Finally, sensors data can view on internet browser application using cloud computing.[2]

Sokratis Kartakis, Weiren Yu, Reza Akhavan, and Julie A. McCann entitled "Adaptive Edge Analytics for Distributed Networked Control of Water Systems" This paper presents the burst detection and localization scheme that combines lightweight compression and anomaly detection with graph topology analytics for water distribution networks. We show that our approach not only significantly reduces the amount of communications between sensor devices and the back end servers, but also can effectively localize water burst events by using the difference in the arrival times of the vibration variations detected at sensor locations. Our results can save up to 90% communications compared with traditional periodical reporting situations.[3]

Prasad et al. (2015) the smart Water Quality Monitoring (WQM) device for Fiji using IoT and remote sensing technologies is shown in this article. The Pacific Islands of Fiji require regular collection and analysis of collected data for the water quality monitoring and uploading this data into the server. In order to monitor water quality, the authors have used IoT and remote sensing technologies. The current measurements can be enhanced by remote sensing. During the entire test period, the system has been proved worth by delivering accurate and consistent data using IoT for water monitoring in real-time. The system proposed by these authors also used a GSM module to forward the data to the mobile user via SMS.

Omar Faruq et al. (2017) A water quality monitoring system based on microcontrollers for people living in Bangladesh's outskirts, where safe drinking water is not available, is provided in this paper. The device has been designed with a high degree of accuracy and is sensitive to several water parameters such as temperature, turbidity and hydrogen potential. (pH) displayed on the LCD monitor. Finally, in this paper, each of the parameter values is compared with the predefined equipment, and sensor values and error are calculated.

2015 IEEE Design of water management system(Author F Ntambi, C P Kruger, B J Silva, G P Hancke)The system consist of 3 wireless sensor sub-system. All communicate with each other wirelessly and send information to gateway connected to a computer which hosts the GUI. Due to wireless transfer of data sometimes delivery of data is not ensured. There are chances of loss of data.

2016 IEEE Smart water management using IoT (Author Sayali Wadekar, Vinayak Vakare, Ram Ratan Prajapati) Water level sensor will provide the level of water present in the water tank and according to the level of water, water motor will automatically turn ON and OFF. Data is displayed on android application. No quality monitoring is performed, so even if water is available in tank, without performing quality check, water will be supplied. The application needs to be downloaded and updated from time to time.

2017 IJIRSET An IoT based model for smart water distribution with quality monitoring (Author Joy Shah) The paper focuses on water distribution using water flow sensor and water control valve will help in even distribution of water and provide adequate amount of water. The model does not use water level sensor, so the availability of water in the tank will not be known. People will not be aware of unavailability of water.

CHAPTER 3

THEORY

OBJECTIVE:

Our other objective is to design a Smart Water Quantity Meter to detect the water level and an IoT based automatic system for switching on the power supply to refill the tank and for switching the power supply off after the threshold level is reached. This is optional and only meant for those households which use underground water or donot have the facility of automation in their homes unlike those living in aprtments and flats already having this facility.

PROPOSED SYSTEM:

The proposed system uses four sensors which are pH, turbidity, ultrasonic, DHT-11, microcontroller unit as the main processing module and one data transmission module ESP8266 Wi-Fi module (NodeMCU). The microcontroller unit is a significant part of the system developed for water quality measurement because The Arduino Mega consumes low power, and it is a small size, where the size is a good use for a crucial point-of-sale technology criterion. Among four sensors, two of the sensors collect the data in the form of analog signals; the MCU has an on-chip ADC that translates the sensor analog signals into the digital format for further study. So, to get this analog output from the sensor, the sensor's analog output of will be connected to the MCU's analog pins. Whereas the other two sensors output directly connected to the digital pins of the MCU units. All the sensors data processed by the MCU and updated to the ThingSpeak server using the Wi-Fi data communication module ESP8266 (NodeMCU) to the central server (Daigavane & Gaikwad, 2017). The block diagram of the system proposed for water quality measurement is shown in Figure .

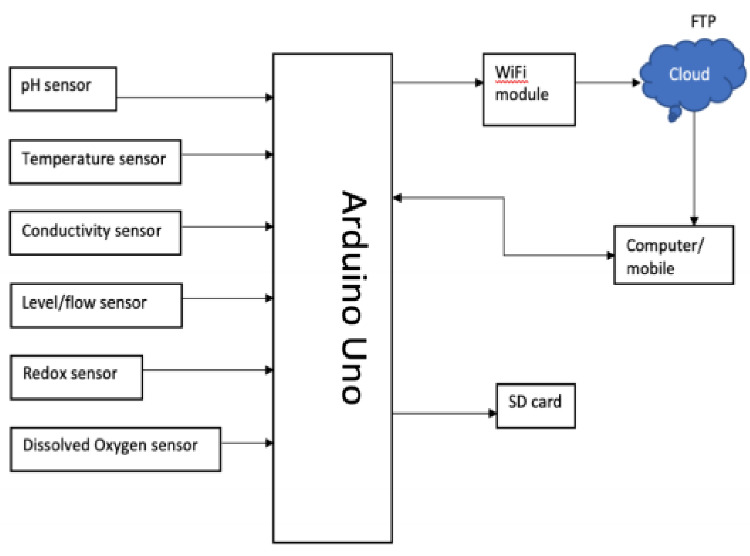


Fig 3.1: Block Diagram of Smart water Monitoring System

IMPLEMENTATION:

A) ARDUINO UNO:

Arduino is an open-source electronics platform based on easy-to-**use** hardware and software. **Arduino** boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.

The **Arduino Uno** is a microcontroller board based on the ATmega328. It has 20 digital input/output pins (of which 6 can be used as PWM outputs and 6 can be used as analog inputs), a 16 MHz resonator, a USB connection, a power jack, an in-circuit system programming (ICSP) header, and a reset button.

Fig 3.1: Arduino uno

B) WIFI MODULE:

The ESP8266 WiFi Module is a self contained SOC with integrated TCP/IP protocol stack that can give any microcontroller access to your WiFi network. The ESP8266 is capable of either hosting an application or offloading all WiFi networking functions from another application processor.

The ESP8266 is capable of hosting an application or offloading all Wi-Fi networking functions from another application processor. Each ESP8266 module come pre-programmed with an AT command set firmware. The ESP8266 module is an extremely cost effective.

Fig 3.2: WiFi Module

C) pH Sensor:

A pH is an electronic device which is used for measuring the pH level in the water. It consists of three types of probes (i) Glass electrode (ii) Reference electrode (iii) combination of gel electrode. pH is described as the “negative logarithm” of hydrogen ion concentration in water.

$pH=-log[H+]$

A pH meter consists of special probes which are connected to an electronic meter that would display the reading. If the pH level is greater than 7 then it is alkaline in nature, if the pH level is less than 7 then it is acidic in nature, and generally the range of pH is 0-14pH.

Fig 3.3: pH sensor

D) FLOW SENSOR:

This sensor basically consists of a plastic body, a rotor and a sensor. The pinwheel rotor rotates when water / liquid flows through the pipe and its speed will be directly proportional to the flow rate. The Hall Effect sensor will provide an pulse with every revolution of the pinwheel rotor.

Fig 3.4: Flow Sensor

E) TURBIDITY SENSOR:

Turbidity sensors measure the amount of light that is scattered by the suspended solids in water. As the amount of total suspended solids (TSS) in water increases, the water's turbidity level (and cloudiness or haziness) increases. Turbidity sensors are used in river and stream gaging, wastewater and effluent measurements, control instrumentation for settling ponds, sediment transport research, and laboratory measurements.

Fig 3.5: Turbidity Sensor.

F) TEMPETATURE SENSOR:

This sensor is an "integrated circuit sensor". The yield voltage is linearly proportional to the Celsius temperature. The "LM35 sensor" is used in this project because the user cannot convert Kelvin to centigrade temperature. It is not suitable for remote applications and directly measures in Celsius. The applications of the temperature sensor are in the microwave, fridges, household devices, and air conditioners. It measures not only the heat but also measures cold temperature. They are two categories of sensors; they are "contact temperature sensor" and "non-contact temperature sensor".

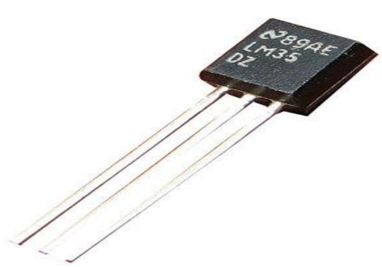


Fig 3.6: Temperature Sensor

G) CLOUD SERVER:

Cloud goes about as a database to store every data generated by the sensors installed in the home. This cloud server help to send email alert about different situation in home to the client.

ThingSpeak is an IoT data collection application for analysis of various sensors, e.g. pH, turbidity, voltage, temperature, moisture, distance, etc. The data collector collects data from edge node devices (this happens with the NodeMCU/ESP8266) and also allows the data to be modified for historical data analysis in a software environment. First, the user must log in with details on his/her server. The channel containing data fields and a status field is the primary component of ThingSpeak activity. After a ThingSpeak channel has been developed, data is modified, processed, and interpreted with MATLAB code, and the data is reacted by tweets and other alerts (Das & Jain, 2017).

CIRCUIT DIAGRAM:

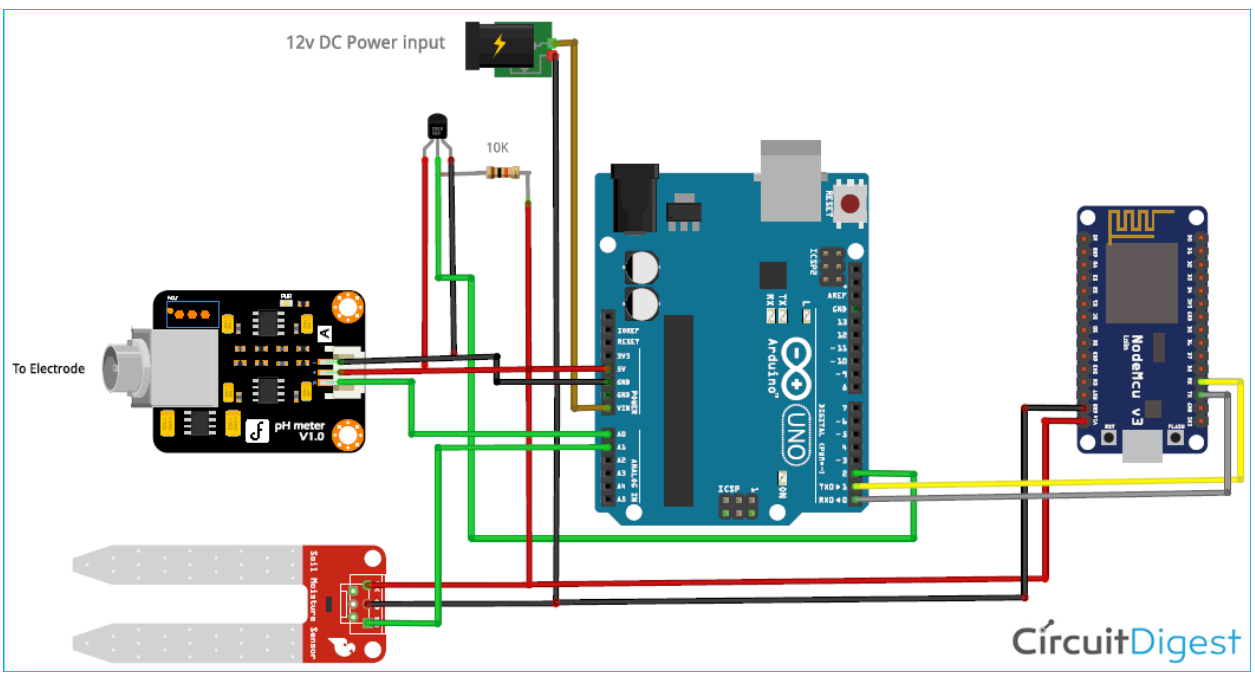


Fig 3.7: Circuit Diagram

FLOW CHART:

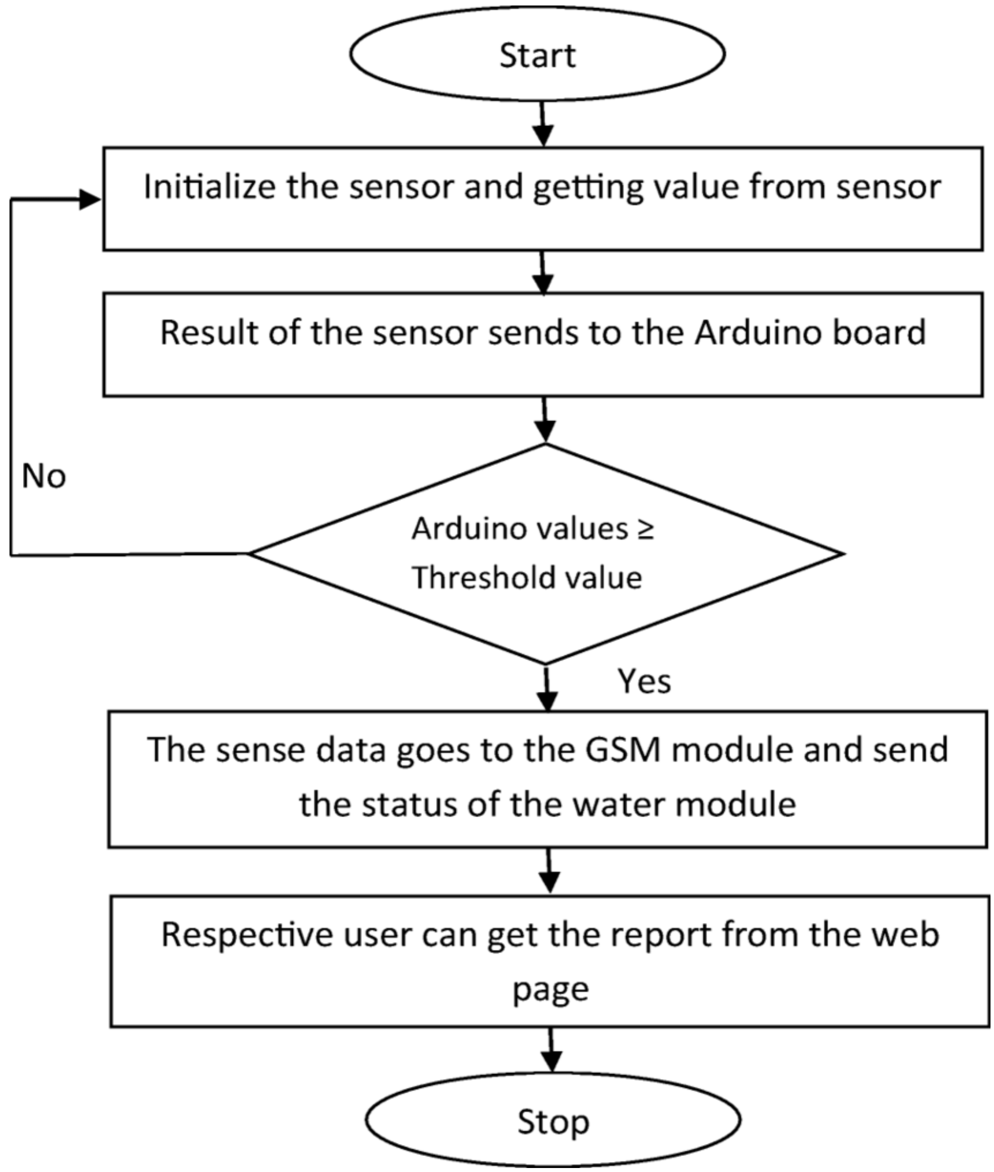
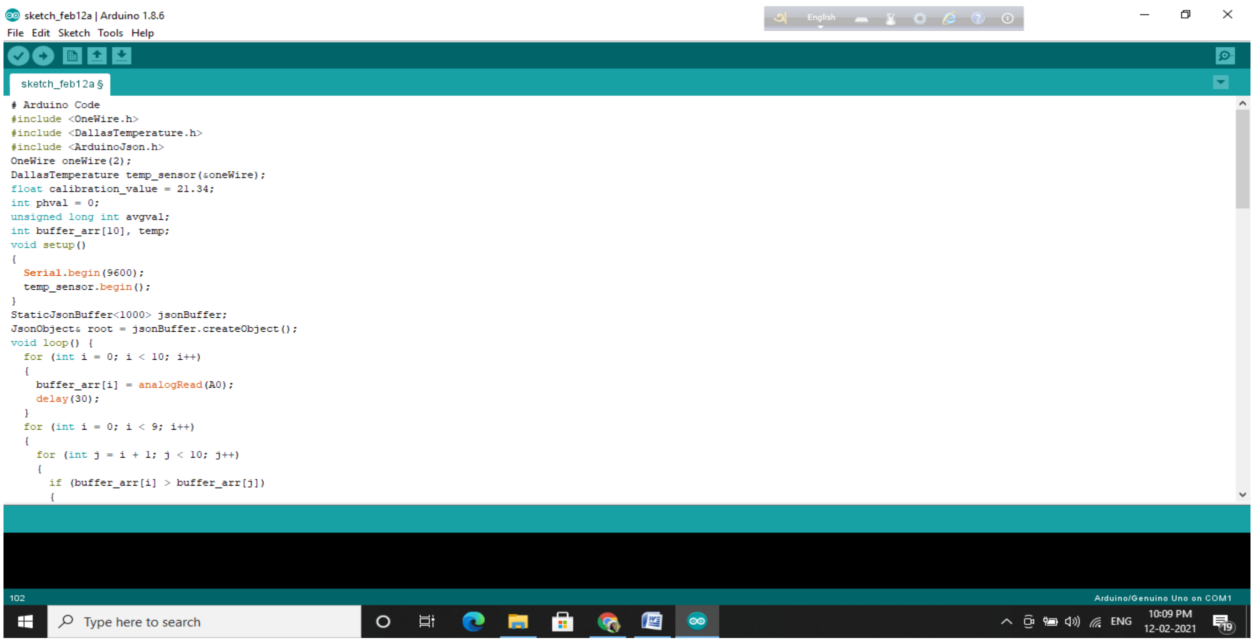
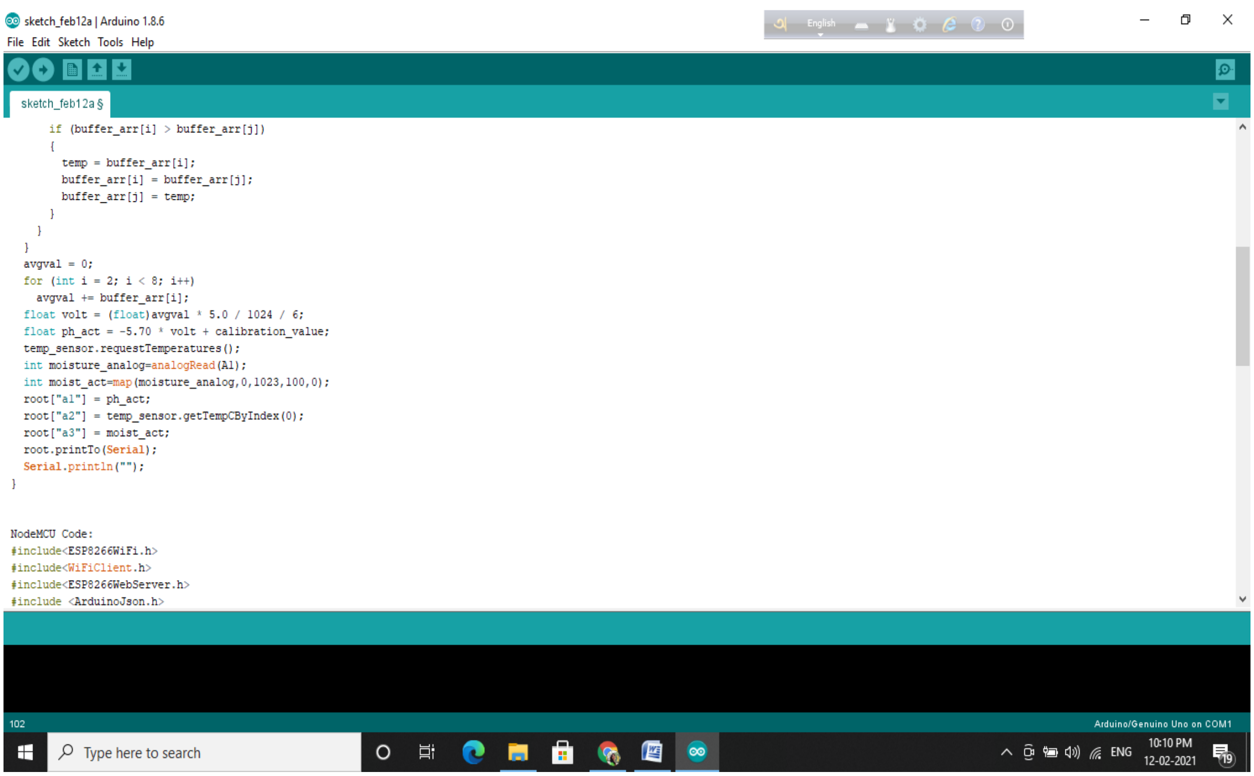


Fig 4.1: Flow Chart of water monitoring system

CODE:



```
sketch_feb12a | Arduino 1.8.6
File Edit Sketch Tools Help
sketch_feb12a §
# Arduino Code
#include <OneWire.h>
#include <DallasTemperature.h>
#include <ArduinoJson.h>
OneWire oneWire(2);
DallasTemperature temp_sensor(&oneWire);
float calibration_value = 21.34;
int phval = 0;
unsigned long int avgval;
int buffer_arr[10], temp;
void setup()
{
  Serial.begin(9600);
  temp_sensor.begin();
}
StaticJsonBuffer<1000> jsonBuffer;
JsonObject& root = jsonBuffer.createObject();
void loop() {
  for (int i = 0; i < 10; i++)
  {
    buffer_arr[i] = analogRead(A0);
    delay(30);
  }
  for (int i = 0; i < 9; i++)
  {
    for (int j = i + 1; j < 10; j++)
    {
      if (buffer_arr[i] > buffer_arr[j])
      {
102
Arduino/Genuino Uno on COM1
Type here to search
ENG 10:09 PM 12-02-2021
```



```
sketch_feb12a | Arduino 1.8.6
File Edit Sketch Tools Help
sketch_feb12a §
      if (buffer_arr[i] > buffer_arr[j])
      {
        temp = buffer_arr[i];
        buffer_arr[i] = buffer_arr[j];
        buffer_arr[j] = temp;
      }
    }
  }
  avgval = 0;
  for (int i = 2; i < 8; i++)
    avgval += buffer_arr[i];
  float volt = (float)avgval * 5.0 / 1024 / 6;
  float ph_act = -5.70 * volt + calibration_value;
  temp_sensor.requestTemperatures();
  int moisture_analog=analogRead(A1);
  int moist_act=map(moisture_analog,0,1023,100,0);
  root["a1"] = ph_act;
  root["a2"] = temp_sensor.getTempCByIndex(0);
  root["a3"] = moist_act;
  root.printTo(Serial);
  Serial.println("");
}


NodeMCU Code:
#include<ESP8266WiFi.h>
#include<WiFiClient.h>
#include<ESP8266WebServer.h>
#include <ArduinoJson.h>
102
Arduino/Genuino Uno on COM1
Type here to search
ENG 10:10 PM 12-02-2021
```

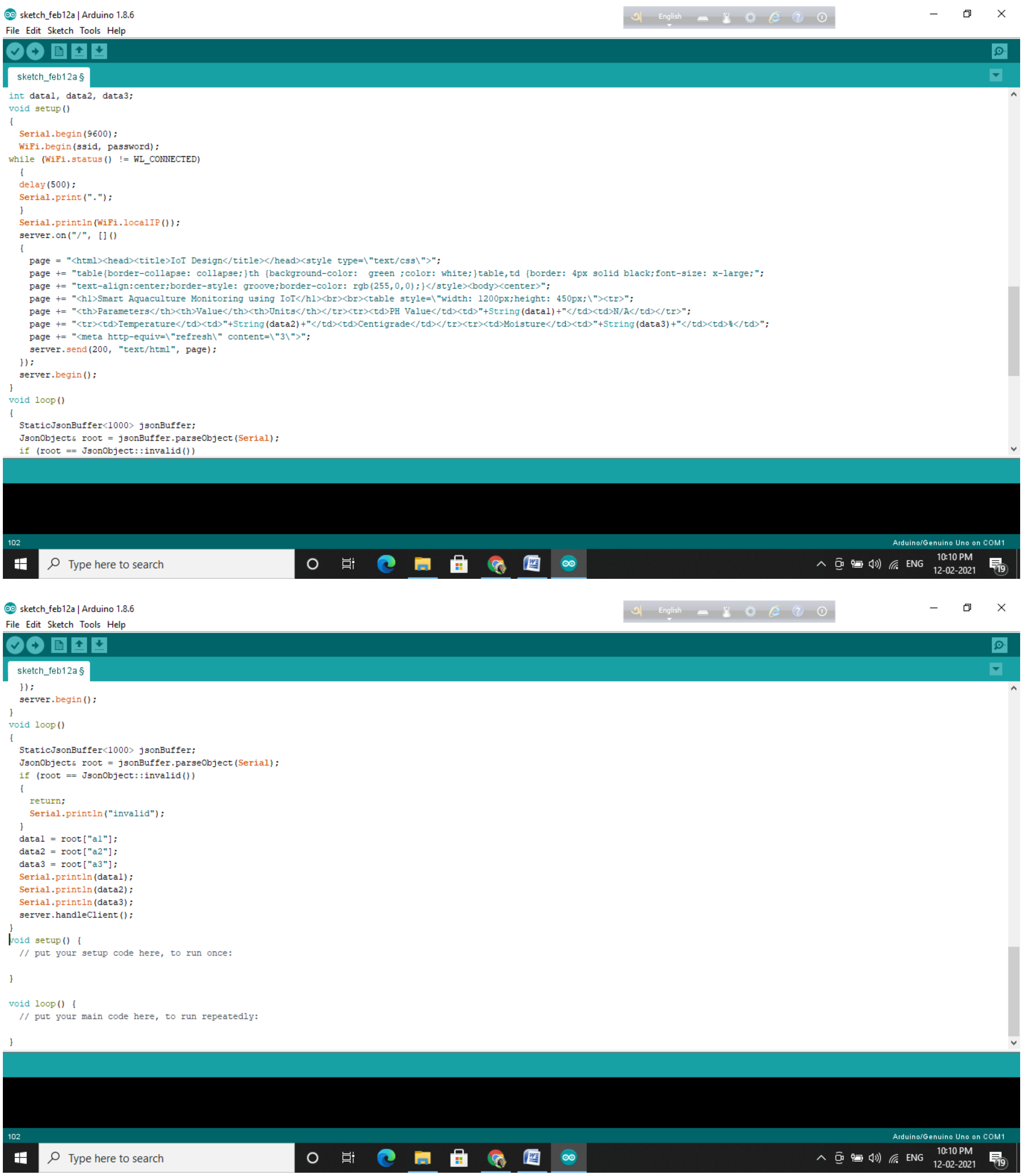


Fig 4.3: IDE CODE

CHAPTER 4

EXPERIMENTAL RESULTS

SIMULATION:

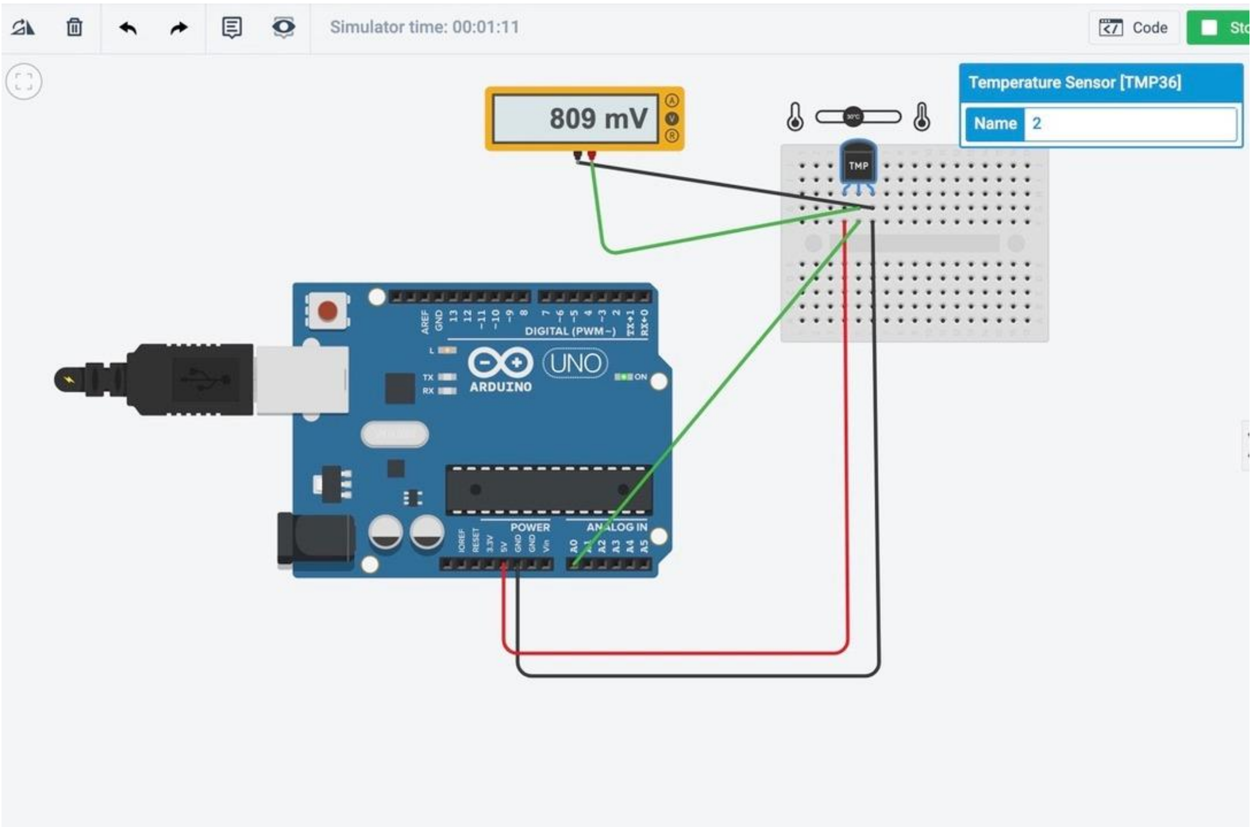


Fig 4.2: Simulation Showing response of temperature sensor

Experimental Table: Values taken for the entire range of TMP36 sensor(-40 deg C to 125 deg C) (later the negative temperatures will be neglected)

Temperature(deg C)	Voltage(mV)
-40	99.9
-31	200
-16	350
-6	450
0	498
2	519
12	619
20	701
30	825
32	829
55	1050
80	1300
92	1430
105	1560
125	1750

Graphical Representation:

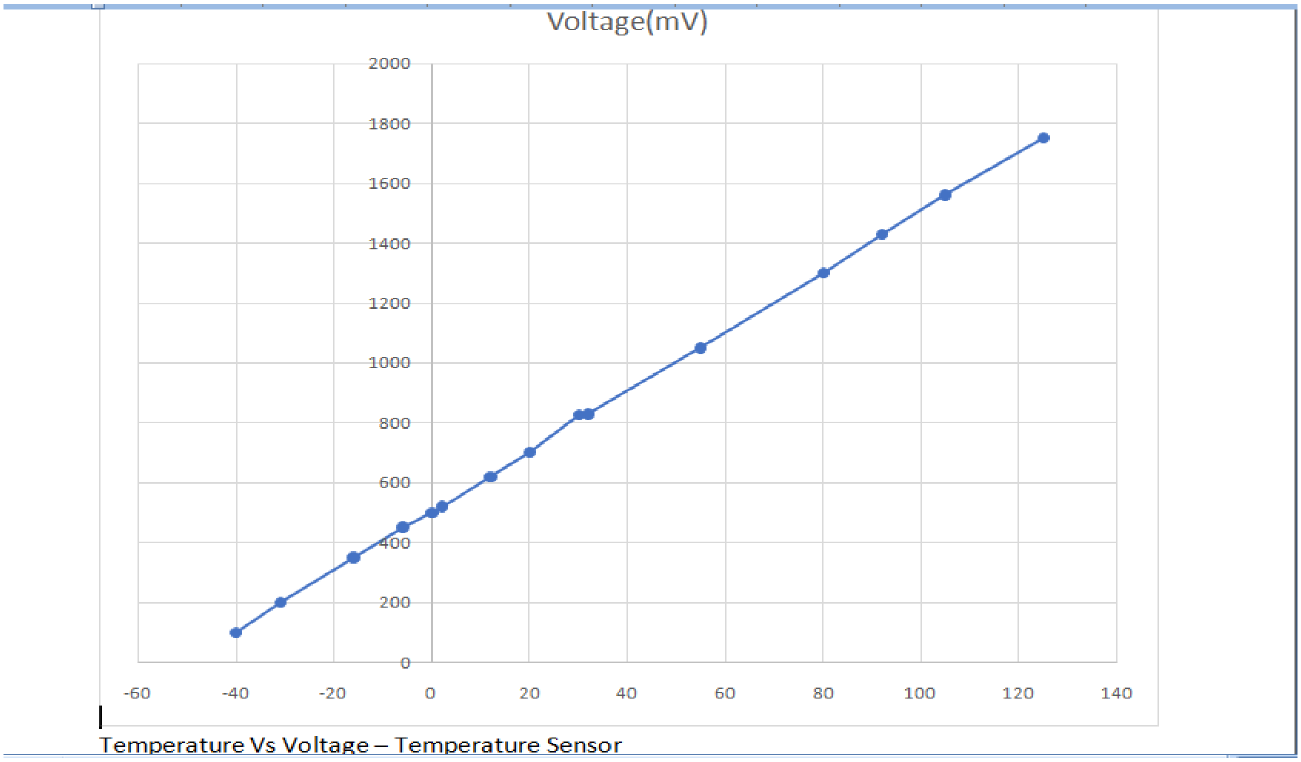


Fig 4.3: Temperature Vs Voltage Curve

Here, we have obtained the response curve of TMP36 sensor. The voltages obtained for Temperatures ranging for 20°C to 30°C will be considered and when the temperature goes below 20°C or above 30°C(i.e.<701mV and >825mV), the signal will be sent to the GSM module.

CHAPTER 5

FUTURE SCOPE AND CONCLUSION

FUTURE SCOPE:

This project has great scope in the long term since water is the key ingredient of life and the pollution of this non-renewable resource will soon reach its peak. Every food industry, agriculture sectors, drinking water industries and suppliers need high quality water because consumption of low quality water and food prepared with low quality water will have a direct impact on health. If this project turns out successful we will try to increase the parameters by addition of multiple sensors. • We wish to help the agricultural fields in the future, because if we maintain the water quality used in the agricultural field that will make the crop and vegetables more healthier. • Further we want to reach other industries requiring good quality water. • If water bill gets implemented in large scale we will include the real time monitoring of units of water consumed in our smart quantity meter.

CONCLUSION:

This research demonstrates a smart water quality monitoring system. Four different water sources were tested within a period of 12 hours at hourly intervals to validate the system measurement accuracy. The results obtained matched with the expected results obtained through research. The temperature relation with pH and conductivity were also observed for all the water samples. GSM technology has been successfully implemented to send alarm based on reference parameter to the ultimate user for immediate action to ensure water quality. Additionally, the parameter references obtained from all the different water sources will be used to build classifiers which will be used to perform automated water analysis in the form of Neural Network Analysis. In a nutshell, the system has proved its worth by delivering accurate and consistent data throughout the testing period and with the added feature of incorporating IoT platforms for real time water monitoring, this should be an excellent contender in real time water monitoring solutions.

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